From Fixed Predation Risk to Co-Evolution: Testing the Emergence of Flocking in an Extended Selfish-Herd Model

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I. Introduction

Predator–prey interactions are inherently adaptive systems in which both sides continuously adjust their behaviour in response to one another. In many social species, prey mitigate predation risk by forming groups such as flocks or herds, giving rise to collective behaviours like alignment, cohesion, and coordinated movement. These behaviours are often interpreted as mechanisms that reduce individual risk through effects such as dilution, confusion, and information transfer. Empirical studies support this view, showing that animal groups can dynamically reorganize under threat, for example through coordinated escape waves or structured information flow during predator encounters. Such observations suggest that collective behaviour is not static, but emerges from ongoing interactions between individuals and their environment Jadhav (2024); Storms et al. (2019) .

A framework for understanding such behaviour is the selfish-herd model Yang (2024), which explains how individual-level selection can give rise to group-level patterns. Recent work has extended this idea using agent-based simulations and evolutionary game theory Yang (2024), demonstrating that a wide range of collective patterns can emerge depending on the distribution of predation risk across positions within a group. In this model, the prey evolve behavioural strategies that determine transitions between positions such as leader, follower, border, and centre, while predator behaviour is represented as a fixed positional risk distribution. This approach makes it possible to identify evolutionarily stable strategies and systematically analyze how different environmental risk structures influence the emergence of collective motion.

However, a key limitation of this framework is that predation risk is treated as fixed Yang (2024) rather than adaptive, which could be less realistic. In real predator–prey systems, predator behavior is not static, but responds to prey organization, leading to feedback between prey strategies and predation pressure. By modeling predator influence as an externally imposed risk distribution, existing approaches do not capture this co-adaptive dynamic, and therefore may overestimate or mischaracterize the conditions under which flocking behavior emerges or remains beneficial. This raises an important question: do the conclusions drawn from fixed-risk models hold when predator strategies are allowed to evolve alongside prey?

To address this gap, this project will extend the selfish-herd framework by introducing a co-evolutionary predator component. Specifically, this project will replace the fixed positional risk distribution with an adaptive predator strategy that determines attack probabilities over prey positions, allowing predation risk to emerge dynamically during simulation. Then the plan is to compare this co-evolutionary model to the original fixed-risk baseline to evaluate how predator adaptation influences the emergence and stability of flocking behaviour, as well as its impact on prey survival and predator efficiency. The central research question of this project is therefore:

Research Question: How does allowing predator strategies to co-evolve with prey, instead of treating predation risk as fixed, affect the emergence and stability of flocking behaviour and prey survival in the selfish-herd framework?

Goals:

- Extend the selfish-herd model by replacing fixed predation risk with an evolving predator strategy
- Compare a fixed-risk baseline with a co-evolutionary predator–prey model
- Evaluate how predator adaptation influences the emergence and stability of flocking behaviour
- Assess the impact of flocking on prey survival (individual and population level) and predator efficiency

II. Related Work

A. Fixed Predation Risk Models

Prior work has extensively studied the emergence of collective behaviour under simplified or fixed predation conditions. For instance, Demšar and Bajec (2015) and Demšar et al. (2016) simulated predator attacks on fish schools, evolving composite predator tactics and demonstrating that intermediate cohesion can confuse attackers, while parallel group motion emerges under balanced antagonistic pressures. Similarly, Chakraborty and Guttal (2020) showed that prey swarms achieve optimal survival at intermediate interaction ranges, where selfish-herd effects dilute individual predation risk.

While these studies provide important insights into how flocking and grouping behaviours can improve survival, they typically treat predation pressure as fixed or only partially adaptive. As a result, the predator–prey interaction is effectively reduced to a static or externally imposed environment. This limits the ability of such models to capture feedback between prey behaviour and predator adaptation. In contrast, our project introduces a co-evolutionary extension in which predator strategies evolve alongside prey, allowing predation risk to emerge dynamically rather than being predefined.

B. Collective Escape and Empirical Flocking

Empirical studies further demonstrate that collective behaviour plays a critical role in predator avoidance. For example, Storms et al. (2019) analyzed starling flocks under falcon predation and observed coordinated escape patterns such as expansion waves and rapid threat propagation, which reduce individual exposure to predators. Similarly, Jadhav (2024) studied sheep flocks responding to herding dogs, showing structured front-to-back information flow that enables coordinated group movement under threat.

These findings highlight that real-world flocking behaviour is highly dynamic and shaped by continuous interaction between predators and prey. However, such studies primarily focus on short-term behavioural responses and do not model the evolutionary processes that give rise to these patterns. Our work builds on these observations by incorporating evolutionary mechanisms, enabling us to study how flocking behaviours emerge and persist over time when both predator and prey strategies are subject to adaptation.

C. Co-evolutionary Predator-Prey Dynamics

Co-evolutionary approaches have been used to study how predator and prey adaptations influence collective behaviour over time. For instance, Yang (2017) combined agent-based simulations with evolutionary principles to investigate how alignment, speed, and cohesion interact under predation pressure, revealing trade-offs that shape group structure. Additionally, Carmichael and Heininger (2015) framed predator–prey interactions within a Red Queen framework, showing that mutual adaptation can lead to bounded evolutionary dynamics rather than unbounded escalation.

While these studies demonstrate the importance of co-adaptation, they often do not directly connect co-evolutionary dynamics to the selfish-herd framework or to models where predation risk is explicitly represented as a positional distribution. In contrast, our project bridges this gap by extending a fixed-risk selfish-herd model into a co-evolutionary setting. Specifically, we replace the static predation risk with an evolving predator strategy that determines risk over prey positions, enabling a direct comparison between fixed-risk and adaptive scenarios. This allows us to evaluate how predator adaptation influences the emergence and effectiveness of flocking behaviour.

III. Approach

To address the research question, an agent-based evolutionary framework that will extend the selfish-herd framework, where predator-prey co-evolution will be implemented. The approach is designed to enable a direct comparison between a fixed-risk baseline, as used in prior work, and a co-evolutionary extension in which predator behaviour adapts alongside prey.

A. Model Design

Two model settings will be considered where, in the baseline setting, the original selfish-herd framework will be re-implemented. In the extended setting, this fixed risk distribution will be replaced with an adaptive predator strategy. Specifically, predators are represented by a parameterized targeting distribution over the same positional categories, which determines the probability of attacking prey at each position. This ensures that predation risk is no longer externally imposed, but emerges dynamically from predator behaviour.

Both prey and predator strategies are updated over successive generations using an evolutionary process, allowing behaviours to adapt in response to interaction outcomes. This enables co-evolutionary dynamics, where predator strategies shape predation risk while prey behaviour simultaneously adjusts to minimize individual risk.

In order to ensure a strong comparison with the original model, the flocking and the evolutionary setup follows the same structure Yang (2024). Particularly, the prey strategies are evolved using genetic algorithms over generations and the prey are evaluated using a utility function based on the inner product of the positional frequency Yang (2024). In the proposed extension, the utility of predators is computed on the basis of the number of prey caught during one iteration, as this is their ultimate goal.

B. Representation of Strategies

Prey strategies will be defined in accordance with the original model, where individuals control their movement behaviour through transition probabilities between positions (e.g., leader to follower, follower to leader, border to leader). This representation captures the balance between attraction and repulsion forces underlying flocking behaviour.

Predator strategies are defined as a vector of weights over prey positions, which is normalized to form a probability distribution. These weights determine how strongly predators target different positions within the group. By evolving these weights, predators effectively adapt their attack preferences in response to prey organisation, thereby shaping the emergent risk landscape.

C. Evaluation Metrics

To assess the effects of predator co-evolution, we evaluate both behavioural and performance outcomes. For prey, we measure mean individual survival time and overall population survival. For predators, we measure capture efficiency. In addition, we analyze the emergence of flocking behaviour through the distribution of individuals across positional categories, allowing us to identify patterns such as cohesive grouping or collective motion.

D. Why This Approach?

This approach is appropriate because it maintains direct comparability with the original selfish-herd model while isolating the effect of predator adaptation. By preserving the same positional framework and modifying only the assumption of fixed predation risk, the proposed extension ensures that any observed differences in flocking behaviour or survival outcomes can be attributed to the introduction of co-evolutionary dynamics. Additionally, the use of an agent-based evolutionary framework allows for the emergence of complex collective behaviour from simple local interactions, making it well-suited for studying adaptive predator–prey systems.

IV. Experiments

Experiments will be conducted under two conditions:

1) a fixed-risk baseline.
2) a co-evolutionary predator–prey model.

By comparing outcomes across these conditions, the effect of predator adaptation on flocking emergence and survival dynamics could be isolated. This controlled comparison will help to determine whether conclusions derived from fixed-risk models remain valid when predation pressure is adaptive.

V. Planning

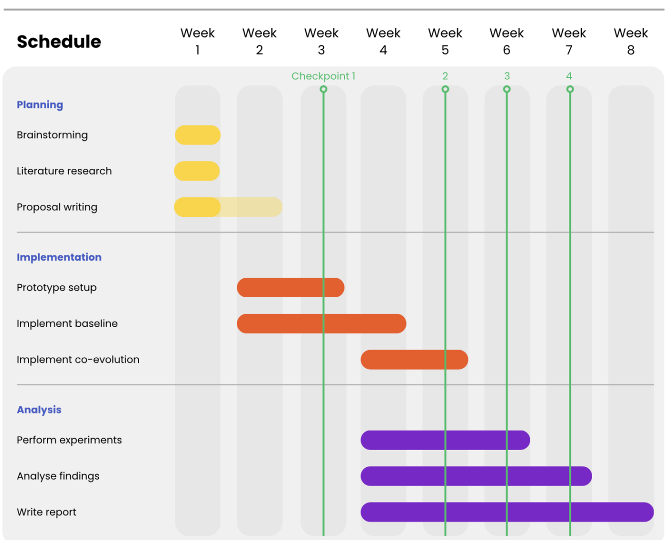

Fig. 1. Proposed schedule for the duration of this project.

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